

WHAM-XREX-PFMG474 Simulator Validation Report

Generated by `python/run_simulator_validation.py`

September 21, 2026

Purpose

Autonomous end-to-end validation of the controller/Transrex-simulator closed loop, per Part C/D of the simulator plan. Each section below covers one test scenario: what was commanded, the expected behavior, the observed result, and a pass/fail verdict. Raw data for every scenario is archived under `shots/` in this repository (gitignored).

1 Closed-loop PID gain sweep ($K_p=0.2, 0.5, 1.0$; $K_i=10, 50, 100$ @ $K_p=1.0$; $K_d=0$ throughout)

Expected: P-only points settle within 500 Hz of $K_p \cdot 40000 / (1 + K_p)$; PI points settle within 500 Hz of the raw 40000 Hz setpoint (integral removes steady-state error), each within 4.0s

Verdict: **FAIL**

Notes: $k_p0.2$: PASS; $k_p0.5$: PASS; $k_p1.0$: PASS; k_i10 : PASS; k_i50 : PASS; k_i100 : FAIL

2 OCP fault mid-shot

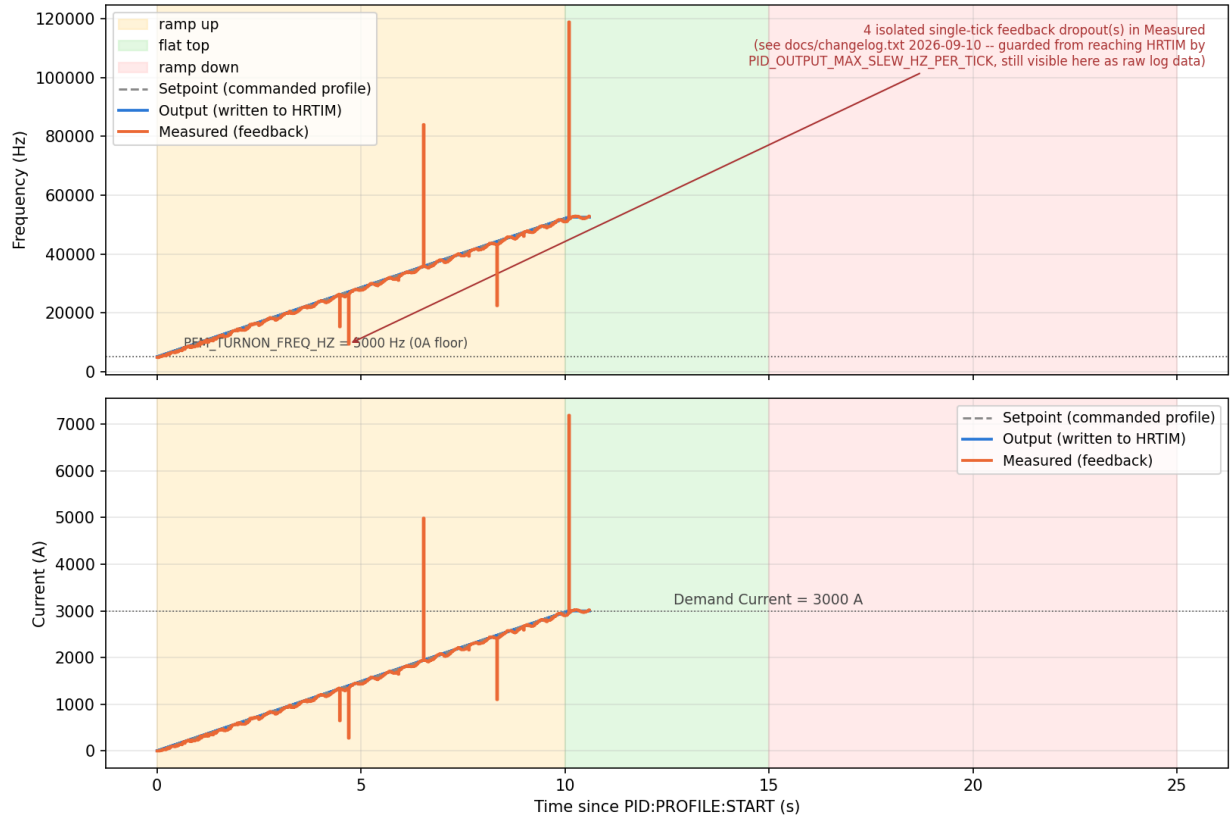
Expected: STATE? reports OVERCURRENT, survivor stays running, visibly derated below its steady-state output

Verdict: **PASS**

Notes: STATE? -> 'OK FAULT OVERCURRENT 1'; survivor ch2 running=1 output=21660; faulted/survivor logged (584/584 samples).

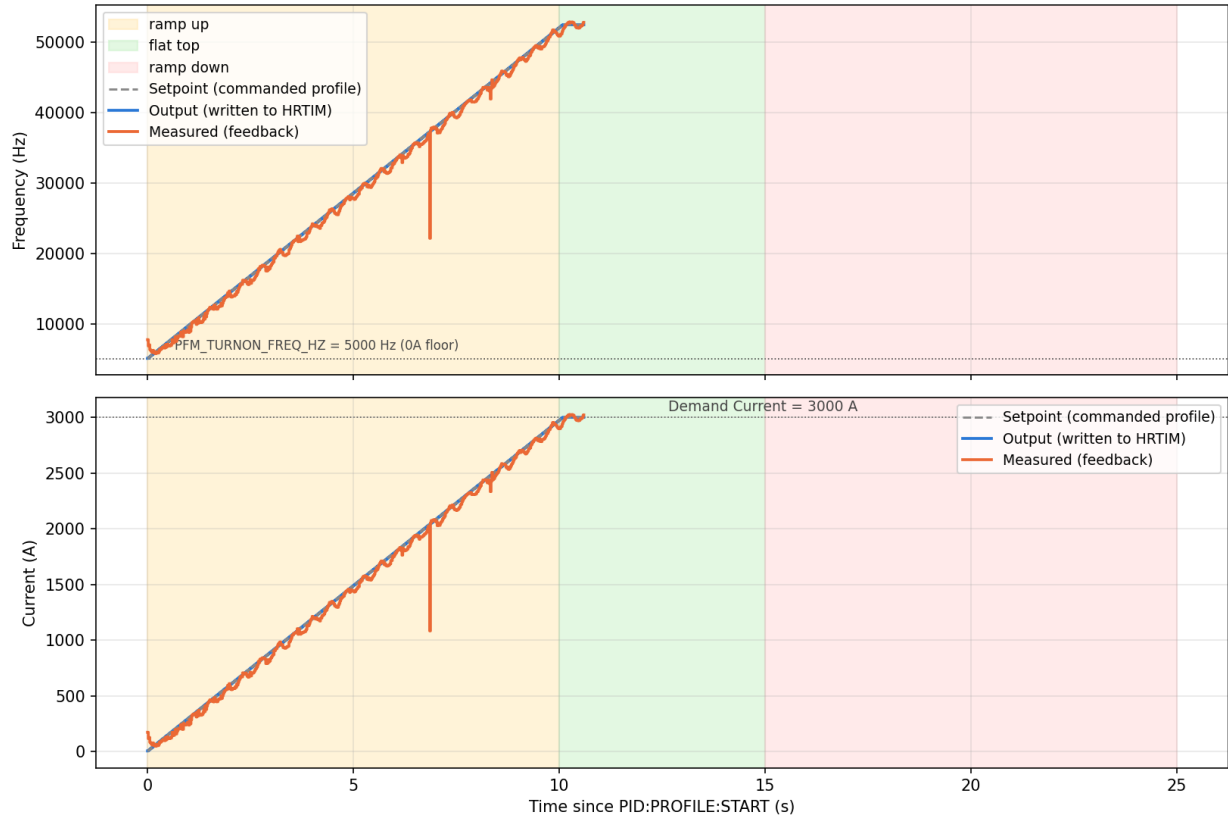
Raw data: `shots/20260921_142918_ocp/ctrl_ch1.csv`

WHAM-XREX-PFMG474 Ch1 -- OK WHAM-XREX-PFMG474 REVA v0.1 d74b0653-dirty
 2026-09-21 14:29:30 | Kp=0 Ki=0 Kd=0
 Ramp=10.00s FlatTop=5.00s Demand=3000A Total=25.00s

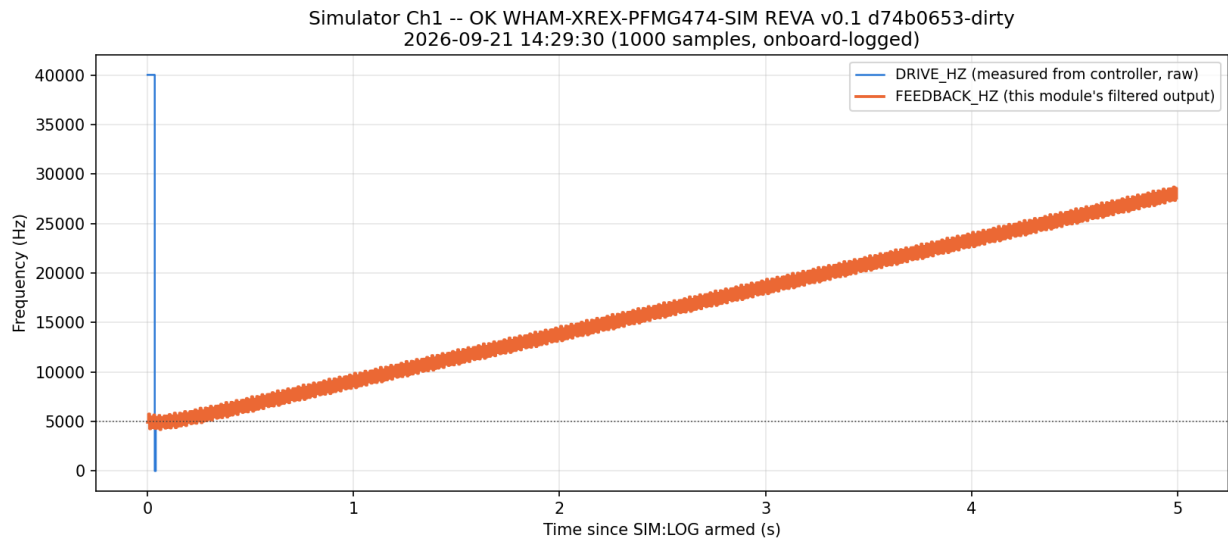


Raw data: shots/20260921_142918_ocp/ctrl_ch2.csv

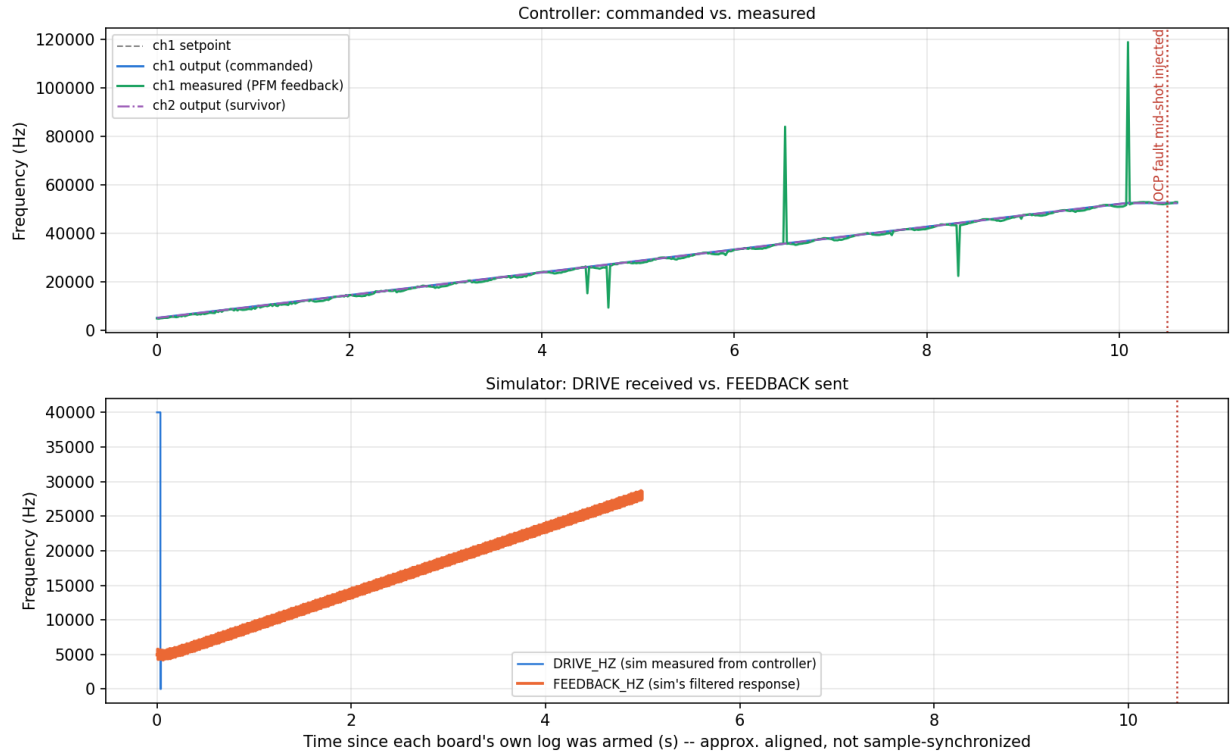
WHAM-XREX-PFMG474 Ch2 -- OK WHAM-XREX-PFMG474 REVA v0.1 d74b0653-dirty
 2026-09-21 14:29:30 | Kp=0 Ki=0 Kd=0
 Ramp=10.00s FlatTop=5.00s Demand=3000A Total=25.00s



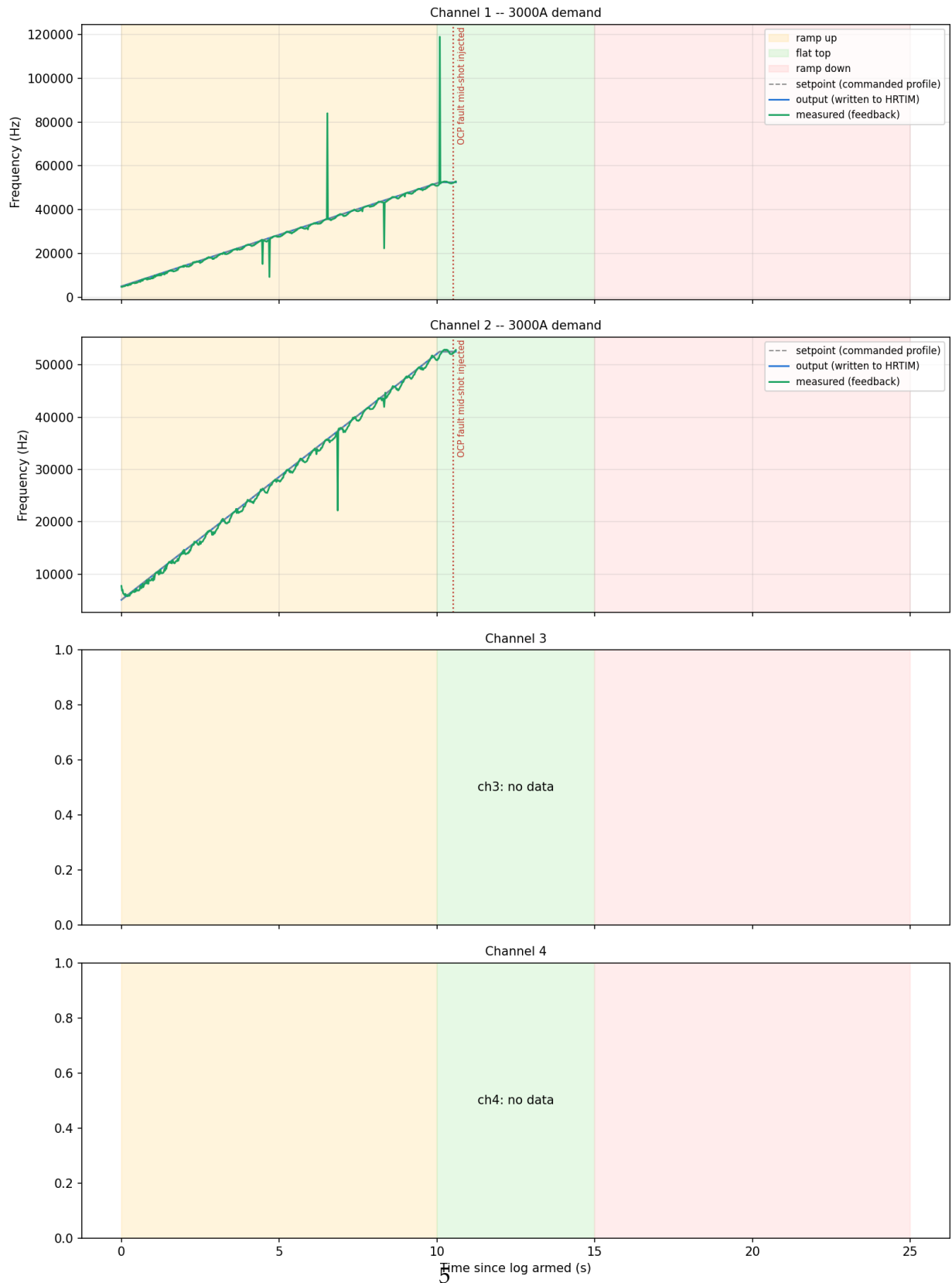
Raw data: shots/20260921_142918_ocp/sim_ch1.csv



ocp -- OCP fault mid-shot
Verdict: PASS | 2026-09-21 14:29:30
STATE? -> 'OK FAULT OVERCURRENT 1'; survivor ch2 running=1 output=21660; faulted/survivor logged (584/584 samples).



ocp -- OCP fault mid-shot
Verdict: PASS
STATE? -> 'OK FAULT OVERCURRENT 1'; survivor ch2 running=1 output=21660; faulted/survivor logged (584/584 samples).



3 Enerpro fault mid-shot

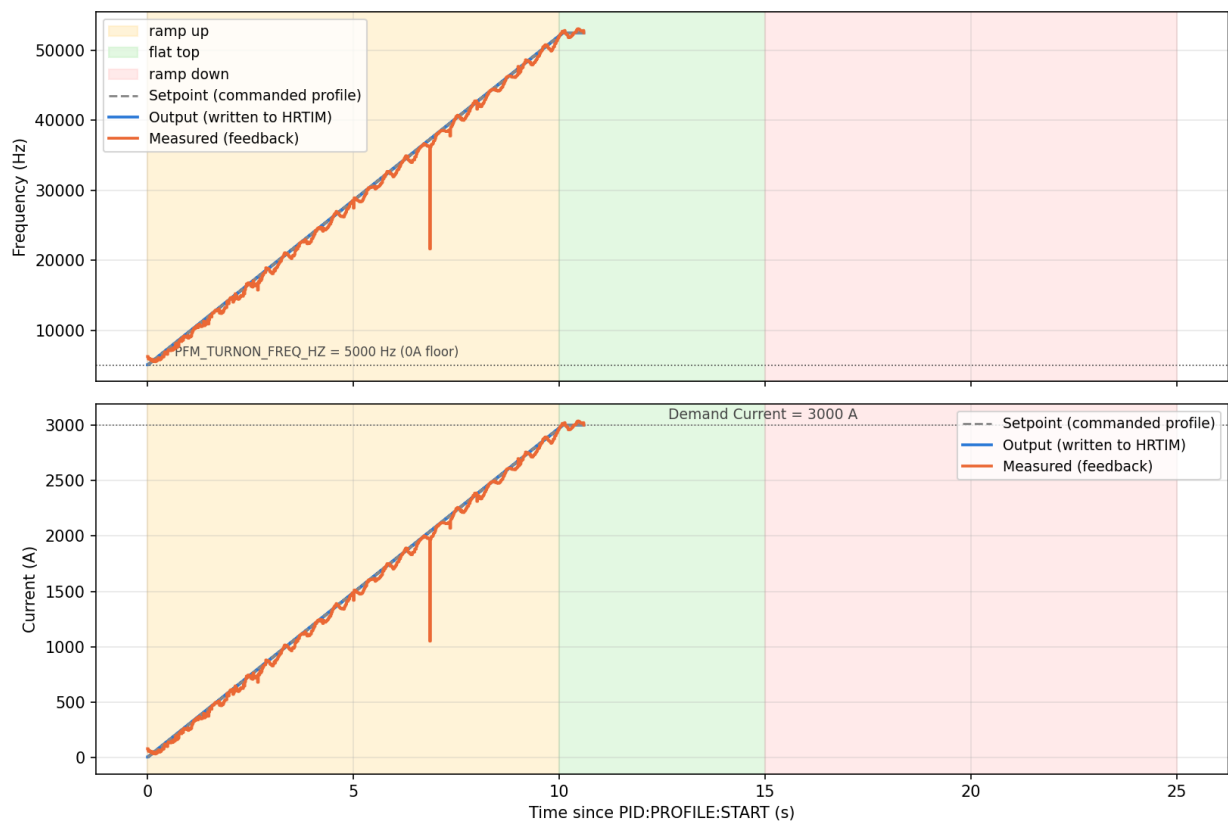
Expected: STATE? reports ENERPRO, survivor stays running, visibly derated below its steady-state output

Verdict: **PASS**

Notes: STATE? -> 'OK FAULT ENERPRO 1'; survivor ch2 running=1 output=21638; faulted/survivor logged (584/584 samples).

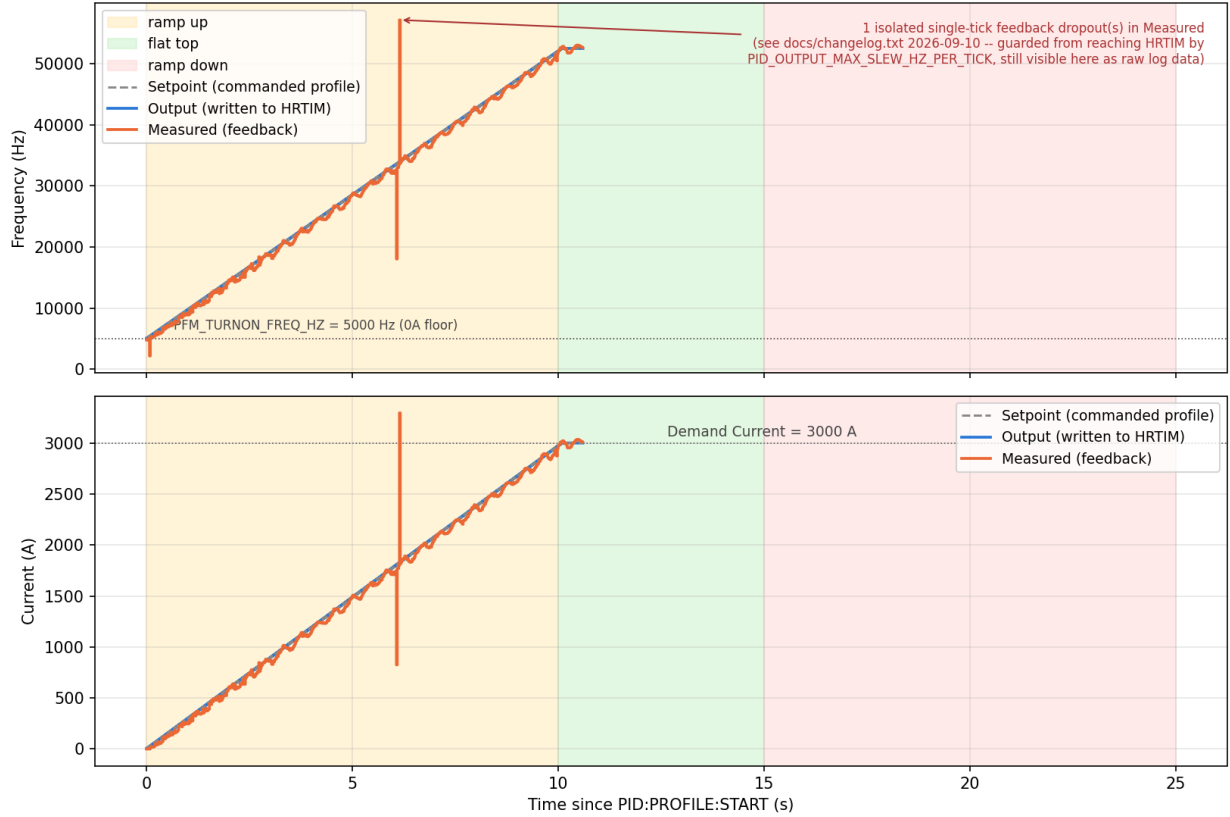
Raw data: shots/20260921_142935_enerpro/ctrl1_ch1.csv

WHAM-XREX-PFMG474 Ch1 -- OK WHAM-XREX-PFMG474 REVA v0.1 d74b0653-dirty
2026-09-21 14:29:48 | Kp=0 Ki=0 Kd=0
Ramp=10.00s FlatTop=5.00s Demand=3000A Total=25.00s



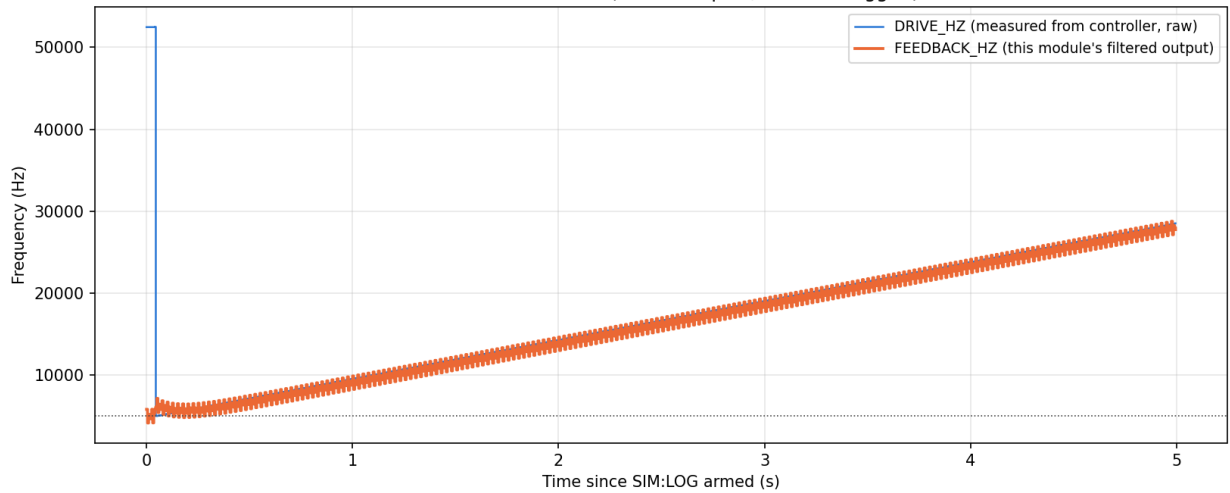
Raw data: shots/20260921_142935_enerpro/ctrl1_ch2.csv

WHAM-XREX-PFMG474 Ch2 -- OK WHAM-XREX-PFMG474 REVA v0.1 d74b0653-dirty
 2026-09-21 14:29:48 | Kp=0 Ki=0 Kd=0
 Ramp=10.00s FlatTop=5.00s Demand=3000A Total=25.00s

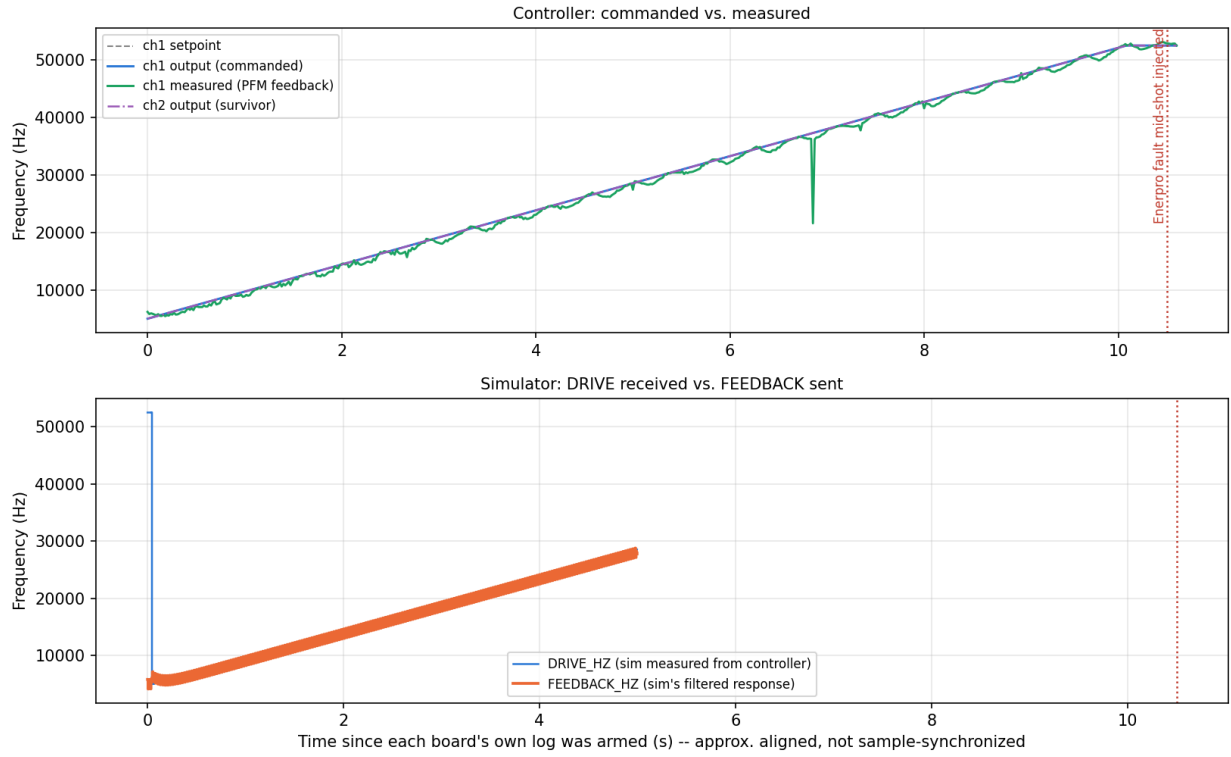


Raw data: shots/20260921_142935_enerpro/sim_ch1.csv

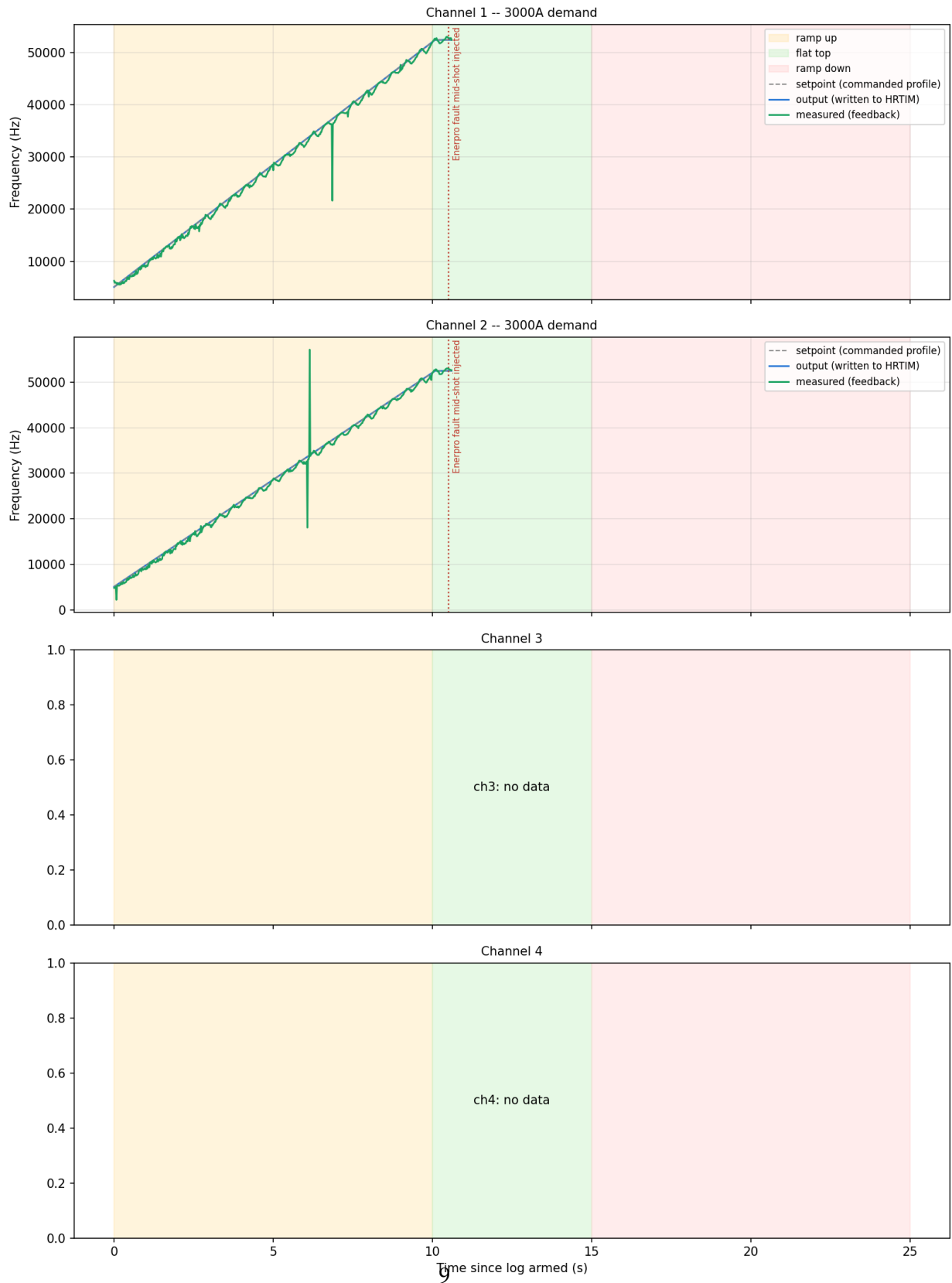
Simulator Ch1 -- OK WHAM-XREX-PFMG474-SIM REVA v0.1 d74b0653-dirty
 2026-09-21 14:29:48 (1000 samples, onboard-logged)



enerpro -- Enerpro fault mid-shot
Verdict: PASS | 2026-09-21 14:29:48
STATE? -> 'OK FAULT ENERPRO 1'; survivor ch2 running=1 output=21638; faulted/survivor logged (584/584 samples).



enerpro -- Enerpro fault mid-shot
Verdict: PASS
STATE? -> 'OK FAULT ENERPRO 1'; survivor ch2 running=1 output=21638; faulted/survivor logged (584/584 samples).



4 Water/Temp fault mid-shot

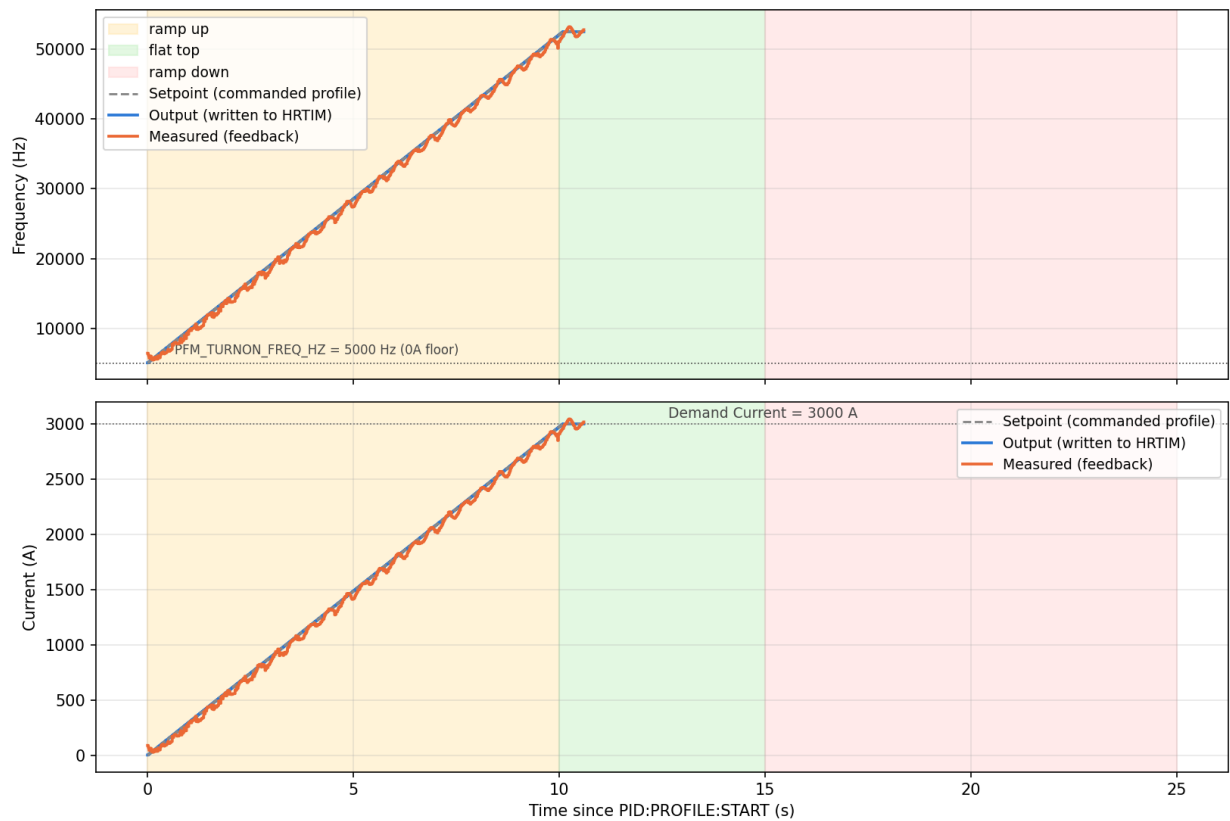
Expected: STATE? reports GENERAL, full system stop expected

Verdict: **PASS**

Notes: STATE? -> 'OK FAULT GENERAL'; survivor ch2 running=1 output=42240; faulted/survivor logged (584/584 samples).

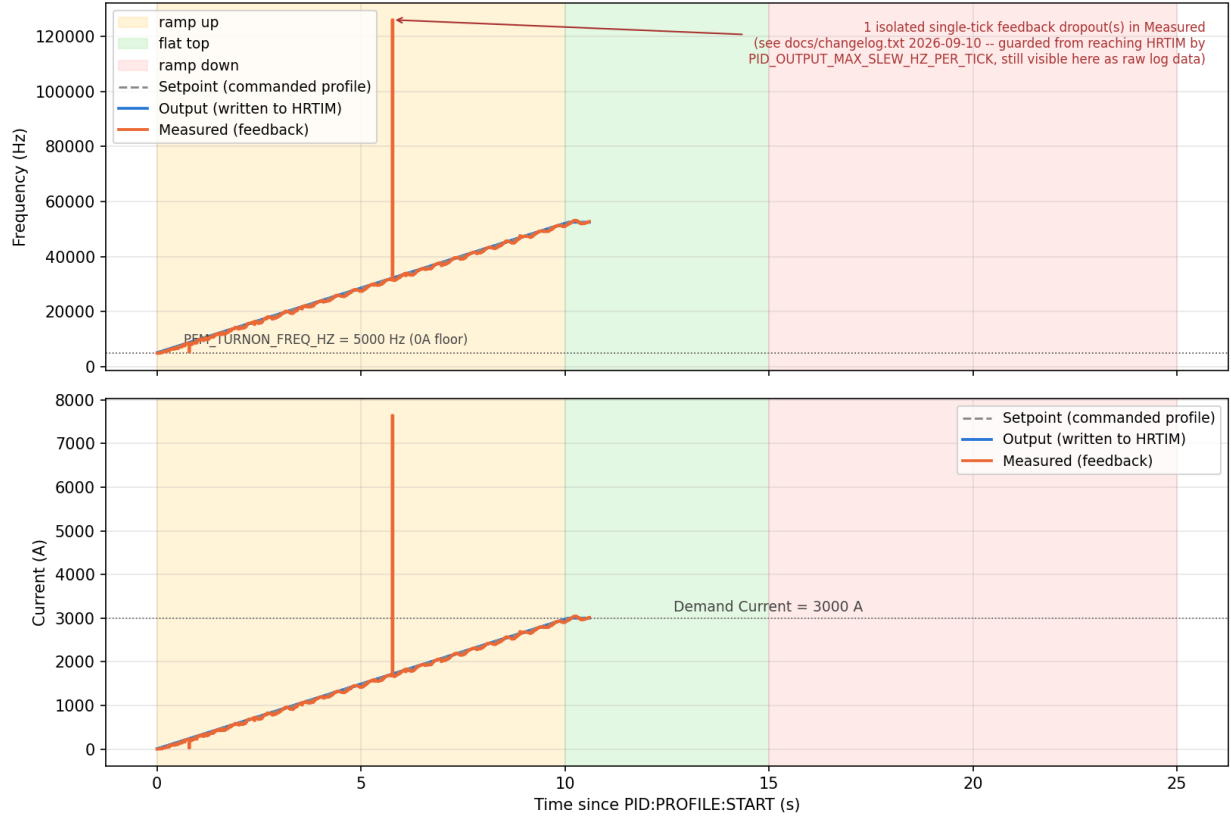
Raw data: shots/20260921_142952_watertemp/ctrl1_ch1.csv

WHAM-XREX-PFMG474 Ch1 -- OK WHAM-XREX-PFMG474 REVA v0.1 d74b0653-dirty
2026-09-21 14:30:05 | Kp=0 Ki=0 Kd=0
Ramp=10.00s FlatTop=5.00s Demand=3000A Total=25.00s



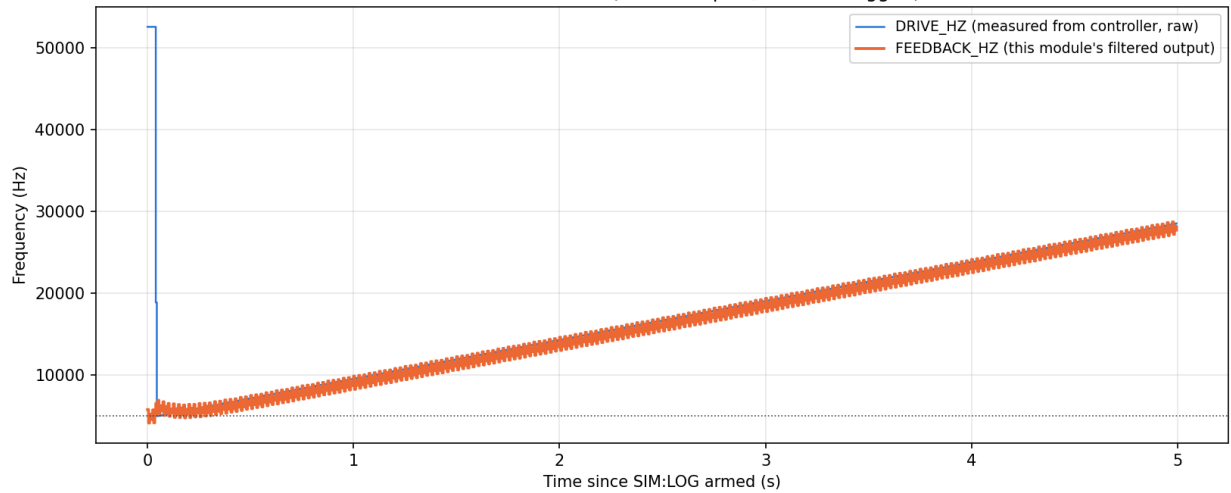
Raw data: shots/20260921_142952_watertemp/ctrl1_ch2.csv

WHAM-XREX-PFMG474 Ch2 -- OK WHAM-XREX-PFMG474 REVA v0.1 d74b0653-dirty
 2026-09-21 14:30:05 | Kp=0 Ki=0 Kd=0
 Ramp=10.00s FlatTop=5.00s Demand=3000A Total=25.00s

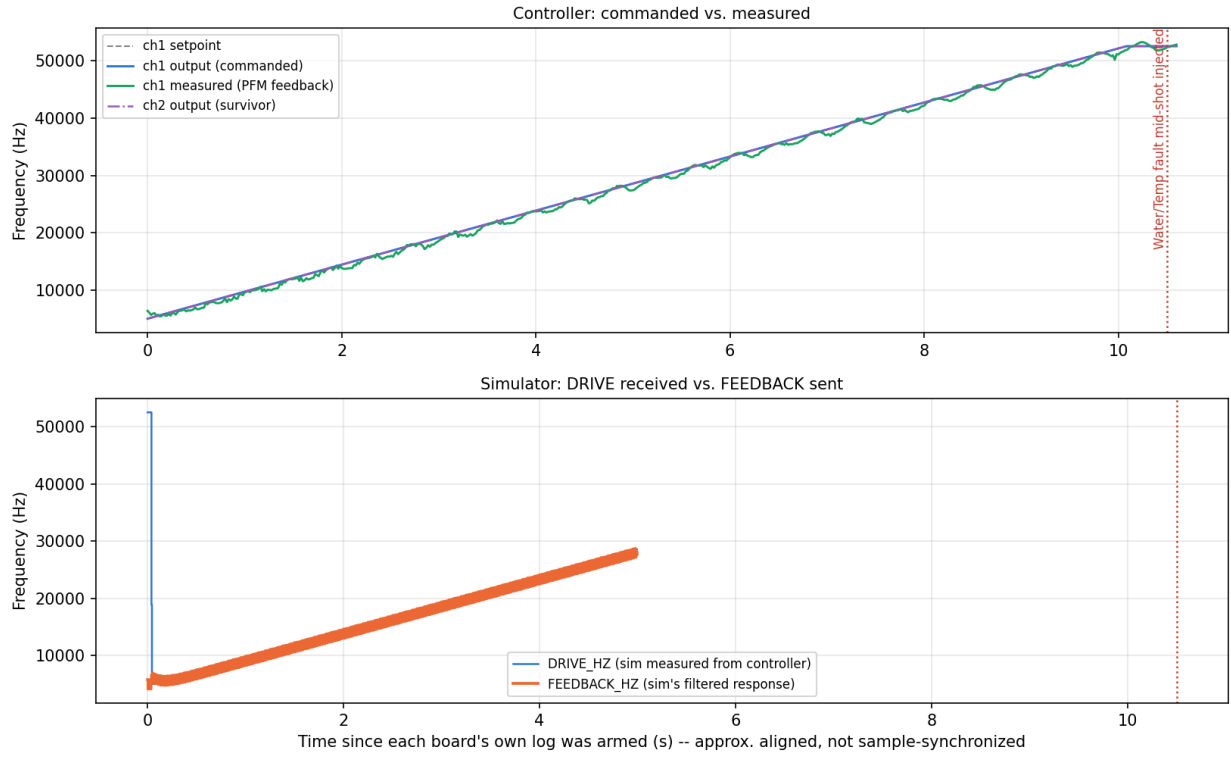


Raw data: shots/20260921_142952_watertemp/sim_ch1.csv

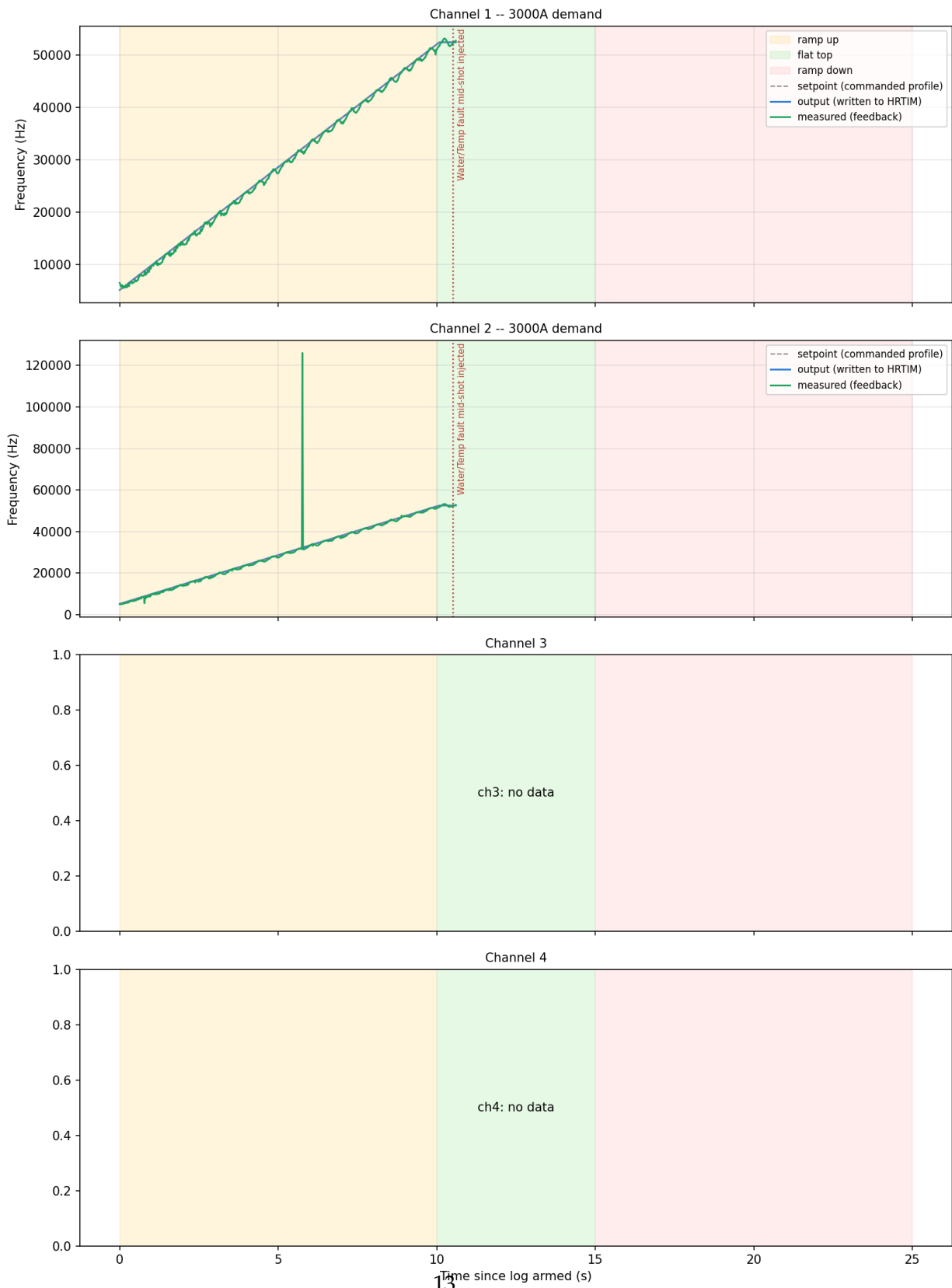
Simulator Ch1 -- OK WHAM-XREX-PFMG474-SIM REVA v0.1 d74b0653-dirty
 2026-09-21 14:30:05 (1000 samples, onboard-logged)



watertemp -- Water/Temp fault mid-shot
Verdict: PASS | 2026-09-21 14:30:05
STATE? -> 'OK FAULT GENERAL'; survivor ch2 running=1 output=42240; faulted/survivor logged (584/584 samples).



watertemp -- Water/Temp fault mid-shot
Verdict: PASS
STATE? -> 'OK FAULT GENERAL'; survivor ch2 running=1 output=42240; faulted/survivor logged (584/584 samples).



5 ENA_OUT/CONTACT_OUT dropped mid-shot

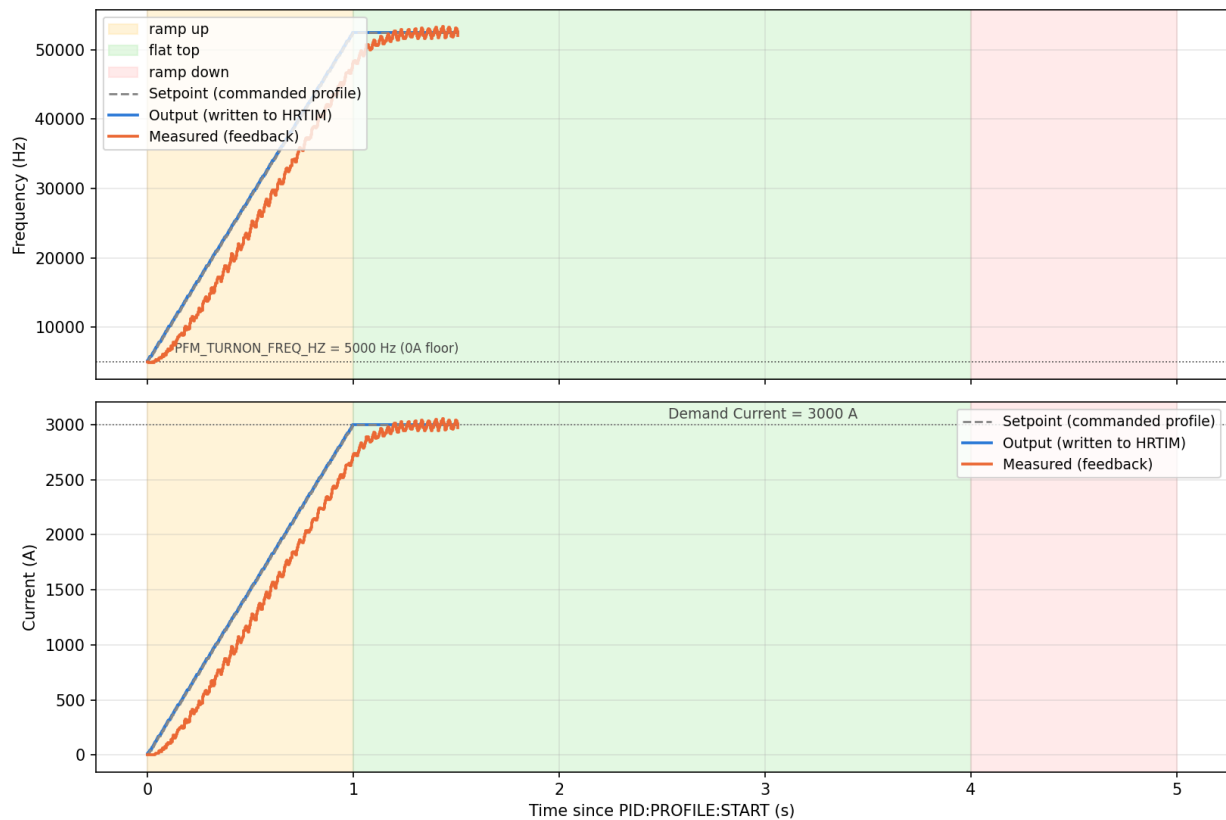
Expected: STATE? reports ENABLE_OUTPUT <ch>, survivor stays running, visibly derated

Verdict: **PASS**

Notes: STATE? -> 'OK FAULT ENABLE_OUTPUT 1'; survivor running=1 output=21660

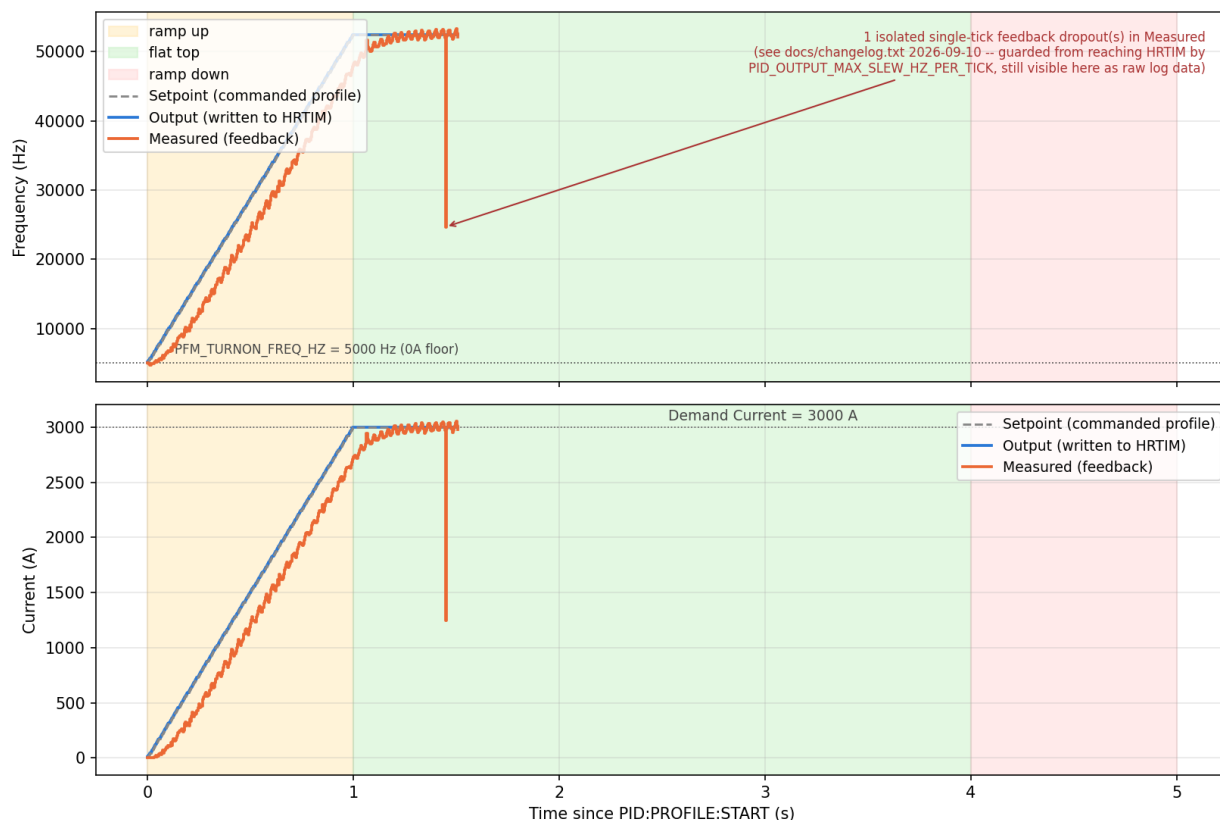
Raw data: shots/20260921_143009_ena_contact/ctrl_ch1.csv

WHAM-XREX-PFMG474 Ch1 -- OK WHAM-XREX-PFMG474 REVA v0.1 d74b0653-dirty
2026-09-21 14:30:13 | Kp=0 Ki=0 Kd=0
Ramp=1.00s FlatTop=3.00s Demand=3000A Total=5.00s



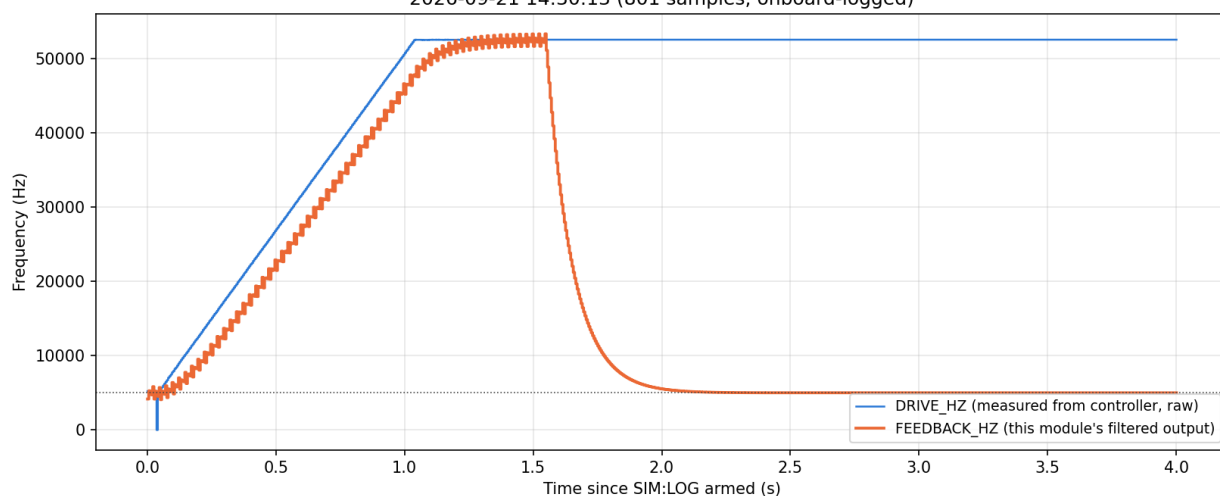
Raw data: shots/20260921_143009_ena_contact/ctrl_ch2.csv

WHAM-XREX-PFMG474 Ch2 -- OK WHAM-XREX-PFMG474 REVA v0.1 d74b0653-dirty
 2026-09-21 14:30:13 | Kp=0 Ki=0 Kd=0
 Ramp=1.00s FlatTop=3.00s Demand=3000A Total=5.00s

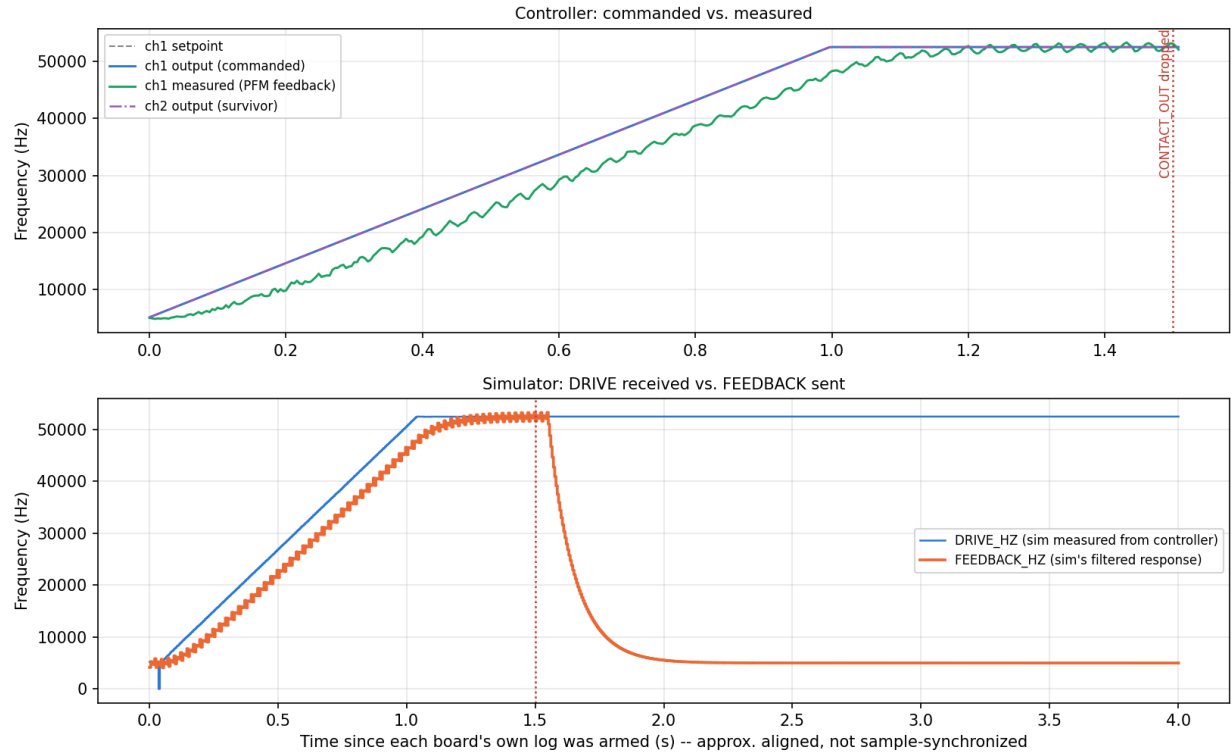


Raw data: shots/20260921_143009_ena_contact/sim_ch1.csv

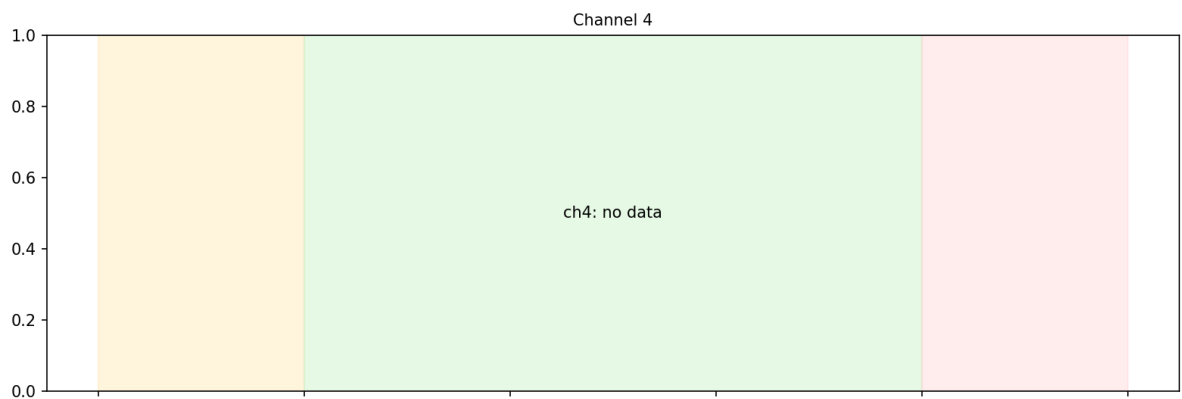
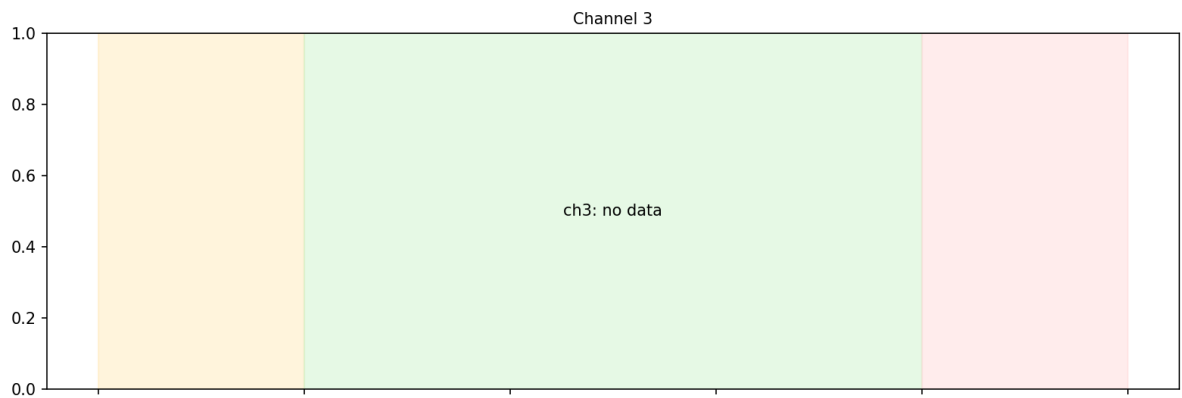
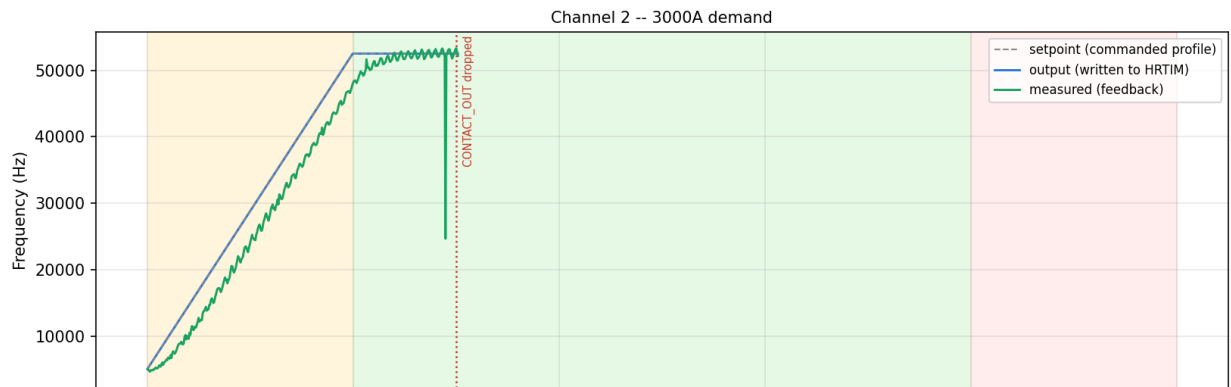
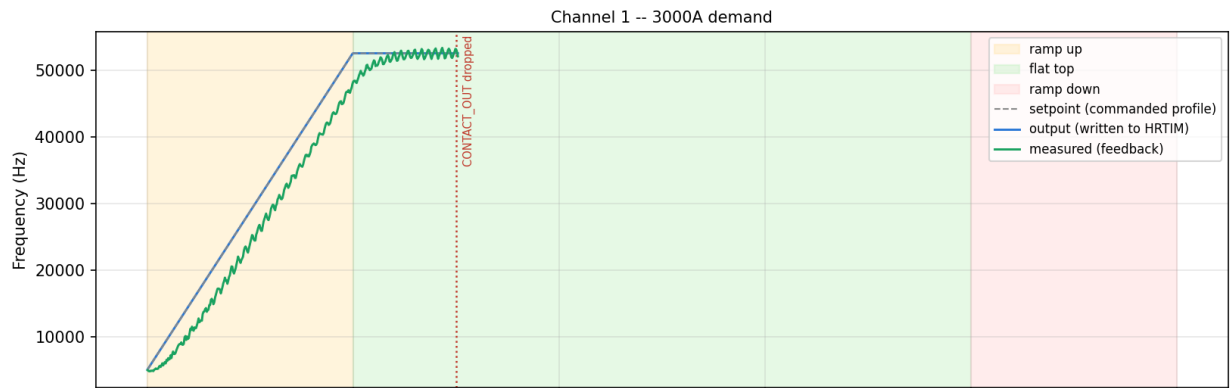
Simulator Ch1 -- OK WHAM-XREX-PFMG474-SIM REVA v0.1 d74b0653-dirty
 2026-09-21 14:30:13 (801 samples, onboard-logged)



ena_contact -- ENA_OUT/CONTACT_OUT dropped mid-shot
Verdict: PASS | 2026-09-21 14:30:13
STATE? -> 'OK FAULT ENABLE_OUTPUT 1'; survivor running=1 output=21660



ena_contact -- ENA_OUT/CONTACT_OUT dropped mid-shot
Verdict: PASS
STATE? -> 'OK FAULT ENABLE_OUTPUT 1'; survivor running=1 output=21660



6 Multi-fault sequencing (OCP then Enerpro, 2 channels)

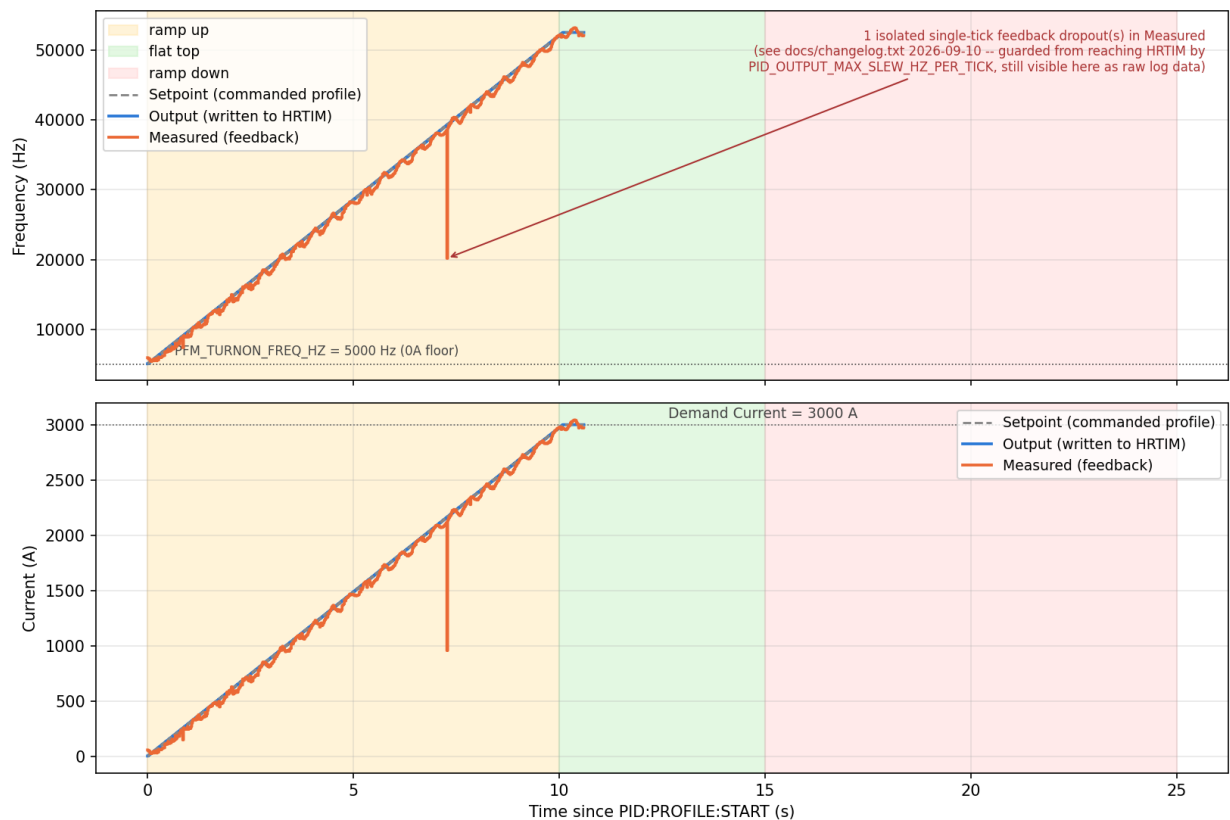
Expected: Both faulted channels (1, 3) end up disabled; STATE? keeps reporting the FIRST fault (OVERCURRENT 1); survivor ch2 stays running and visibly derated below its steady-state output (52500 Hz for 3000.0A) after EACH fault

Verdict: **PASS**

Notes: after 1st ('OK FAULT OVERCURRENT 1'): survivor running=1 output=29989; after 2nd ('OK FAULT OVERCURRENT 1'): survivor running=1 output=24769; ch1 enabled='OK 0'; ch3 enabled='OK 0'

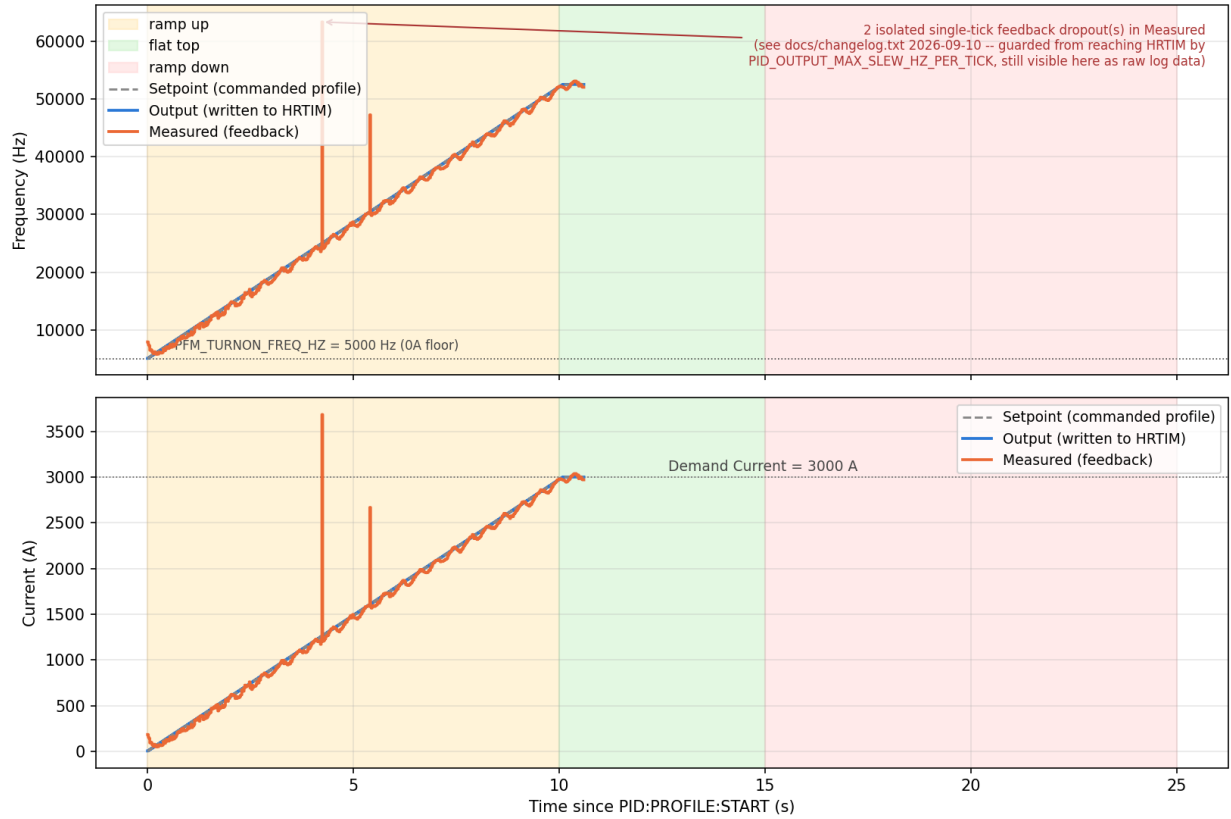
Raw data: shots/20260921_143017_multi_fault/ctrl_ch1.csv

WHAM-XREX-PFMG474 Ch1 -- OK WHAM-XREX-PFMG474 REVA v0.1 d74b0653-dirty
2026-09-21 14:30:28 | Kp=0 Ki=0 Kd=0
Ramp=10.00s FlatTop=5.00s Demand=3000A Total=25.00s



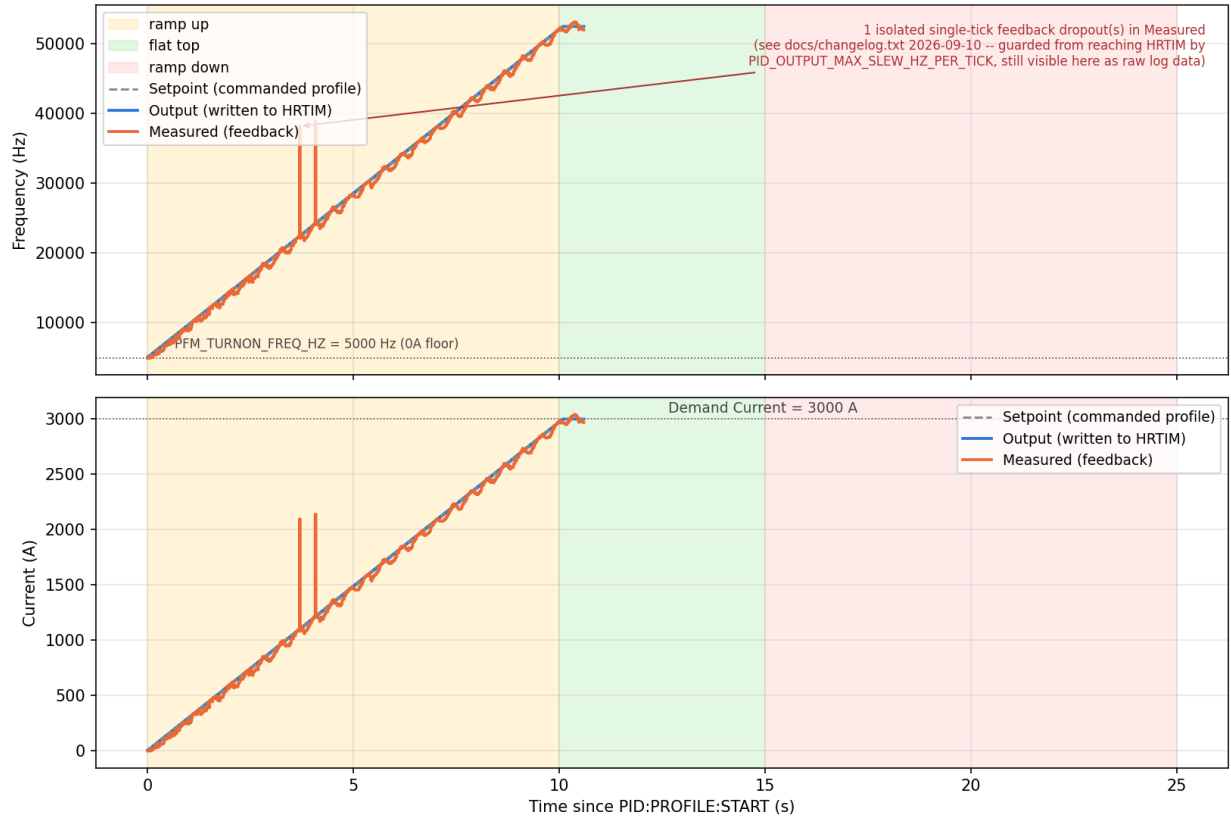
Raw data: shots/20260921_143017_multi_fault/ctrl_ch3.csv

WHAM-XREX-PFMG474 Ch3 -- OK WHAM-XREX-PFMG474 REVA v0.1 d74b0653-dirty
 2026-09-21 14:30:28 | Kp=0 Ki=0 Kd=0
 Ramp=10.00s FlatTop=5.00s Demand=3000A Total=25.00s

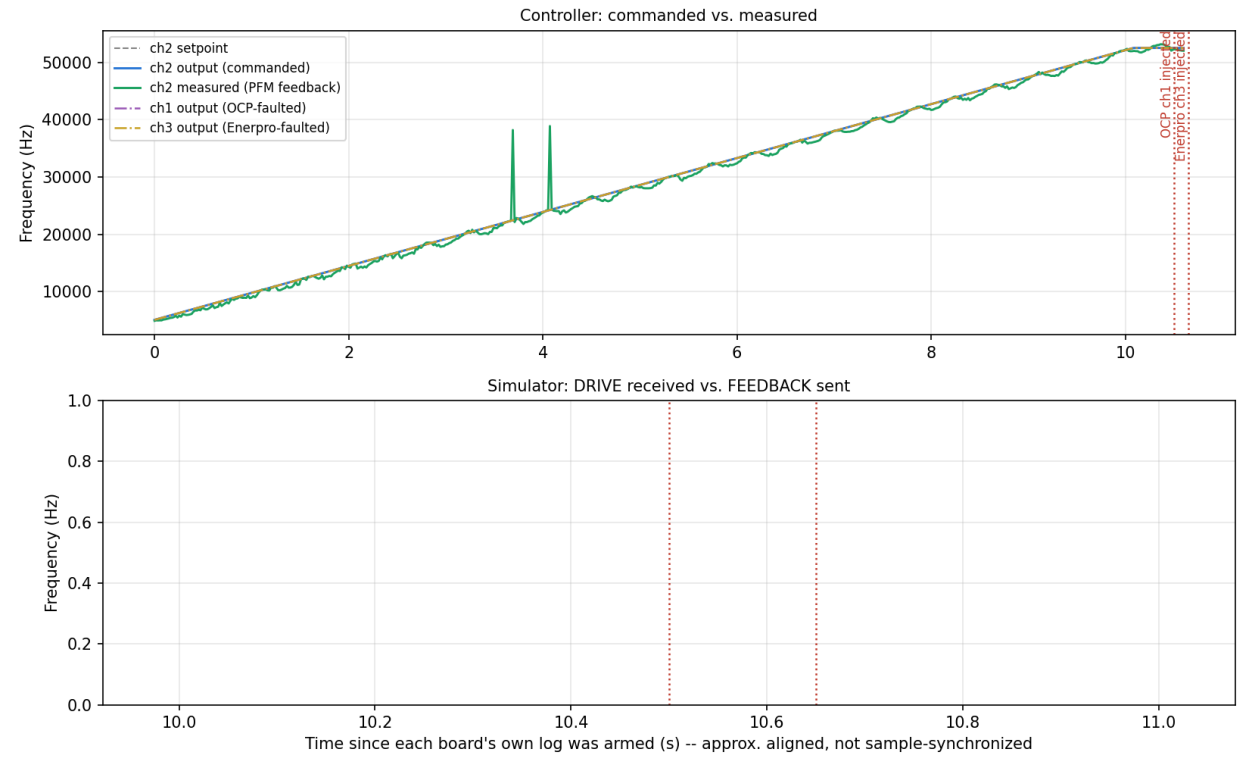


Raw data: shots/20260921_143017_multi_fault/ctrl_ch2.csv

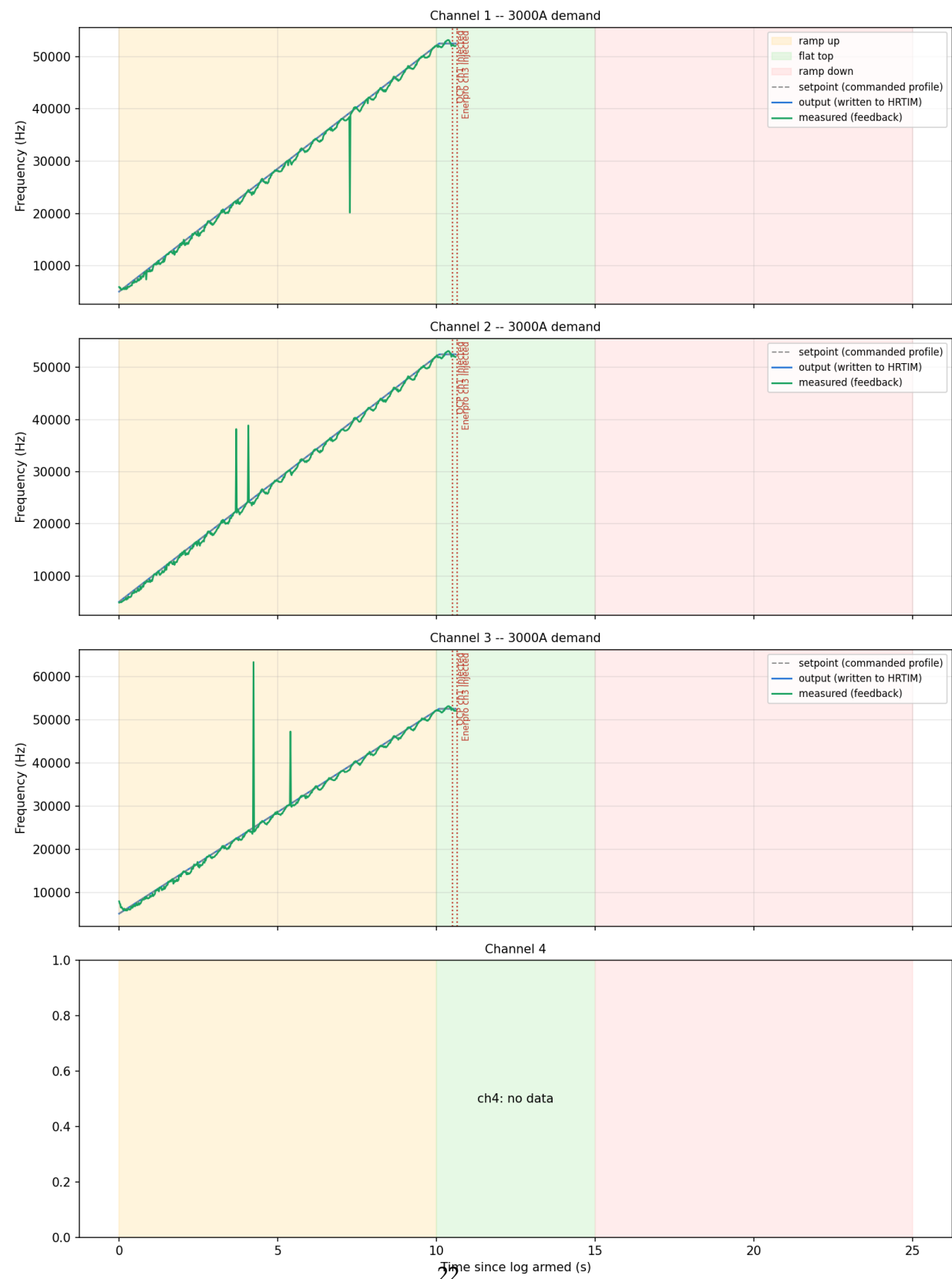
WHAM-XREX-PFMG474 Ch2 -- OK WHAM-XREX-PFMG474 REVA v0.1 d74b0653-dirty
 2026-09-21 14:30:28 | Kp=0 Ki=0 Kd=0
 Ramp=10.00s FlatTop=5.00s Demand=3000A Total=25.00s



multi_fault -- Multi-fault sequencing (OCP then Enerpro, 2 channels)
Verdict: PASS | 2026-09-21 14:30:28
st ('OK FAULT OVERCURRENT 1'): survivor running=1 output=29989; after 2nd ('OK FAULT OVERCURRENT 1'): survivor running=1 output=24769; ch1 enabled



multi_fault -- Multi-fault sequencing (OCP then Enerpro, 2 channels)
Verdict: PASS
LT OVERCURRENT 1'): survivor running=1 output=29989; after 2nd ('OK FAULT OVERCURRENT 1'): survivor running=1 output=24769; ch1 enabled='OK 0'; ch



7 All 4 channels gated and firing simultaneously (600/2000/3600/5400 A)

Expected: Every channel's REAL measured feedback (received across the simulator fiber) tracks its own commanded output within 500 Hz median error during the flat-top (post-ramp steady state), simultaneously, with no cross-channel interference

Verdict: **PASS**

Notes: ch1 flat-top measured_vs_output median error=377Hz; ch2 flat-top measured_vs_output median error=387Hz; ch3 flat-top measured_vs_output median error=362Hz; ch4 flat-top measured_vs_output median error=363Hz

all_channels -- All 4 channels gated and firing simultaneously (600/2000/3600/5400 A)
Verdict: PASS
output median error=377Hz; ch2 flat-top measured_vs_output median error=387Hz; ch3 flat-top measured_vs_output median error=362Hz; ch4 flat-top measured_vs_output median error=362Hz

